

Research paper

New species of Myxidiidae Thélohan, 1892 (Myxosporaea: Bivalvulida) found in characiform fish from the basin of Tocantins River in eastern Brazilian Amazonia

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ABSTRACT

Coelozoic parasites of the family Myxidiidae were observed in fish of the order Characiformes captured in the middle Tocantins River, Maranhão, Brazil, within the transition between the Cerrado savanna biome and the eastern extreme of the Brazilian Amazon Forest. The analysis of the morphological characteristics of the parasites, complemented with a Bayesian phylogenetic analysis, supported the description of new specie *Myxidium imperatrizensis* n. sp., found parasitising the gallbladder of specimens of the fish *Triportheus angulatus*. The study is also only the second report of the occurrence of *Myxidium* parasites in fish of the family Triportheidae in Brazilian rivers. The results of the present study expand the known distribution of the genera *Myxidium* the basin of the Tocantins River, the largest hydrographic basin located entirely within Brazil, which encompasses parts of both the Cerrado and Amazon biomes.

1. Introduction

Myxozoans of the family Myxidiidae Thélohan, 1892 are primarily coelozoic parasites, although a few species are histozoic [1]. Myxidiids infect vertebrate and invertebrate hosts, in both aquatic and terrestrial environments [2,3]. The genus *Myxidium* Bütschli, 1882 is the most diverse myxidiid genus, with more than 232 recognised species [1,4], followed by *Zschokkella* Auerbach, 1909, in which approximately 97 valid species have been described to date [5]. Most of these species are parasites of marine and freshwater fish [6–8]. Phylogenetic analysis of this group, based on the few specimens of *Zschokkella* that present published information on the SSU rDNA gene, revealed four distinct clades, each containing members of this genus and others Myxidiidae, revealing a strong relationship with the hosts' habitat and in the tropism of tissues [5,6].

The most recent review of the myxidiids that parasitise New World fish reported the occurrence of 33 *Myxidium* species and five species of the genus *Zschokkella*. However, only eight *Myxidium* species have been found in Brazilian rivers up to now [9,10], which almost certainly represents a very minor fraction of the true diversity of these genera in tropical South America [2].

With a large variation in relation to the shape of the spore, which varies from spindle-shaped ellipsoidal to slightly curved to sigmoids, the main morphological character used in the differentiation between the genera *Myxidium* and *Zschokkella* is the shape and direction of opening of the polar capsules. While *Myxidium* has polar capsules at or near the end of the spore and which generally open in opposite directions, in *Zschokkella* the polar capsules are, in most cases, almost spherical and slightly subterminal and both cling to one side of the sporadic body [1,5].

The present study describes one new freshwater myxidiid specie that

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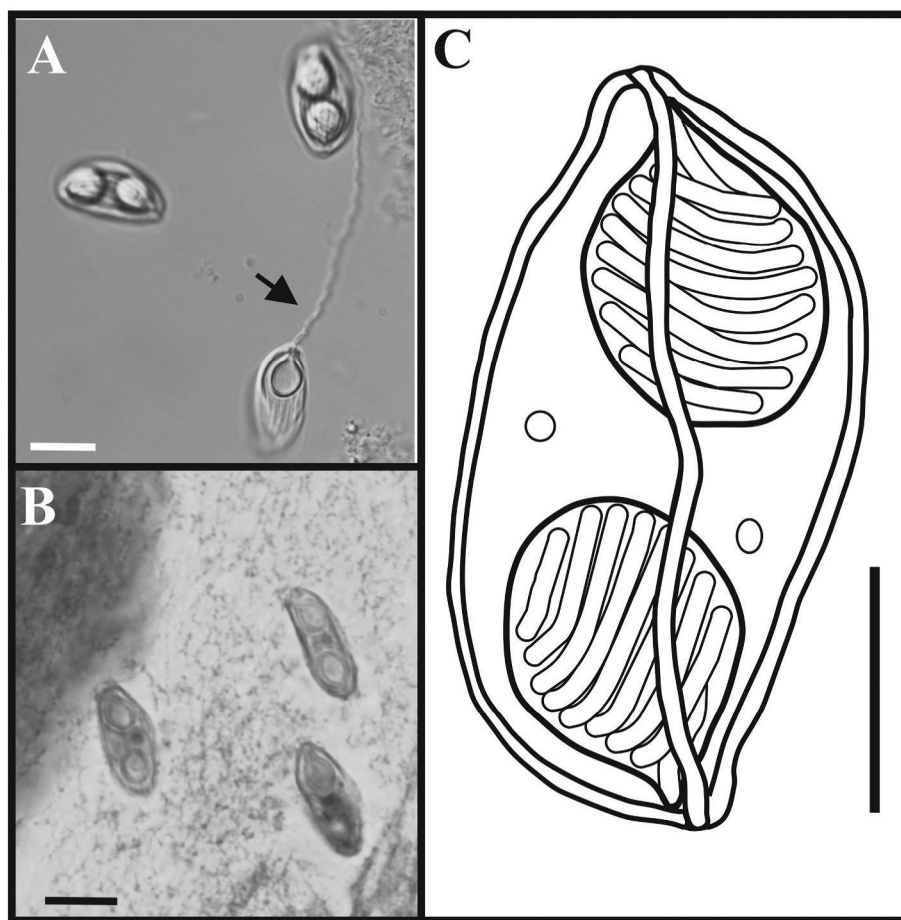


Fig. 1. *Myxidium imperatrizensis* n. sp.: A - Photomicrography in Differential Interference Contrast (DIC) of the fresh spores. Arrowhead is the Extruded polar filament; B - Histological sections of the gallbladder of *Triportheus angulatus*, stained with Hematoxylin-Eosin; C - Line drawing of spores in sutural view. Scale bar 10 μ m.

parasitise the gallbladder of fish of the order Characiformes captured on the Tocantins River, in northern Brazil. The definition of the new specie was based on morphometric data, scanning electron microscopy and partial sequences of the Small Subunit ribosomal ribonucleic acid (SSU rDNA).

2. Material and methods

2.1. Sample collection and morphological analysis

Fish were collected on the Tocantins River between September 2017 and August 2019 using hand-lines, resulting in the collection of 70 *Triportheus angulatus* (Characiformes: Triportheidae). The specimens were collected within the urban zone of two towns, Imperatriz (5.52°S, 47.49°W) and Estreito (6.55°S, 47.45°W), in western Maranhão state, within the eastern extreme of the Amazon Forest biome. Specimen collection was authorised by Brazilian Biodiversity Authorization and Information System (SISBIO 44396–2).

The specimens were transported to the Ecology and Limnology Laboratory of the Maranhão State University of the Tocantins Region, where they were anaesthetised and euthanised for parasitological examination. The experimental procedures were approved by the Ethics Committee for Experimental Research in Animals of the Federal Rural

University of Amazonia (CEUA/UFRA no. 013/2014).

Once the presence of parasites was detected, photomicrographs of the fresh material were obtained by optical microscopy, and by Differential Interference Contrast (DIC) to determine the morphometric parameters of the spores of the parasites found in the gallbladder of the hosts. The morphometry of the myxidiids parasites of freshwater Brazilian fish was compared using a Principal Components Analysis (PCA), run in PAST 3.0 [11,12].

Gallbladders containing spores were fixed in Davidson solution for 24 h and processed for histological sectioning [13]. For Scanning Electron Microscopy (SEM), the contents of the gallbladders containing cysts was fixed for 24 h in 5% glutaraldehyde and post-fixed in 2% osmium tetroxide, dehydrated in an increasing ethanol series, dried to the critical point, and metallised [14] to obtain images using a TESCAN Vega 3 LMU electron microscope in the Ultrastructure Laboratory of the Institute of Animal Health and Production at the Federal Rural University of Amazonia in Belém, Brazil.

2.2. Extraction and sequencing of the DNA

DNA samples from the parasites were extracted using a commercial PureLink® Genomic DNA kit (Invitrogen, EUA), following the manufacturer's specifications. Segments of SSU rDNA were amplified by PCR

using the universal eukaryote primers ERIB1 and ERIB10 in the first amplification [15], and the primers MyxospecF and 18R in the second round of amplification [16,17]. The PCR was run in a final reaction volume of 25 μ l using Master Mix Taq polymerase (PROMEGA, Madison, USA), with the following parameters adjusted for the first amplification cycle: initial denaturation at 94 °C for 5 min, followed by 36 cycles at 94 °C for 30 s, 60 °C for 40 s, and 72 °C for 45 s, with a final extension at 72 °C for 5 min. The second round of amplification was based on initial denaturation at 94 °C for 90 s, followed by 30 cycles at 94 °C for 20 s, polymerisation at 46 °C for 30 s, and extension at 65 °C for 60 s, with a final extension at 65 °C for 10 min.

The PCR products were visualised by electrophoresis in agarose gel and purified using an ExoSAP-IT® PCR Product Cleanup kit (GE Life Sciences), to remove the excess primers and dNTPs. The products were then sequenced using Big Dye Terminator v3.1 (Applied Biosystems, Foster City, CA) in an ABI 3100 sequencer (Applied Biosystems).

2.3. Phylogenetic analyses

The final sequences were assembled in the Codon Code Aligner software (Codon Code Corporation, Dedham, Massachusetts) and compared with data obtained from GenBank, using the Basic Local Alignment Search Tool (BLAST) of the National Center for Biotechnology Information (NCBI). The multiple alignment of the SSU rDNA sequences was run in BioEdit [18], with the ambiguous regions and variable segments with no phylogenetic signal being excluded [19,20].

The substitution models for the Bayesian Inference (BI) were selected using the Bayesian Information Criterion (BIC), run in jModeltest 2.1.10 [19,20,22], which identified the GTR + G model as the most appropriate nucleotide substitution model for the SSU rDNA dataset. The Markov Chain Monte Carlo (MCMC) algorithm was executed in BEAST v.1.8.4 [21,23] with 10,000,000 generations sampled every 10,000 steps [24]. The BI result was used to reconstruct the phylogenetic tree, using a database that included 40 sequences of different species from the families Chloromyxidae, Myxidiidae, Sphaeromyxidae, and Sphaerosporidae. *Buddenbrockia plumatellae* (AY074913) and *Tetracapsuloides vermiformis* (KX013243) were included as an external group.

3. Results

Examination of fish specimens revealed coelocystic gallbladder infections caused by myxozoans of the Myxidiidae family in six of the 70 *Triporthus angulatus* (8.6%). The infections were identified due to the presence of spores floating in the bile fluid and immature forms associated with the internal epithelium of the gallbladder. The morphological characteristics of the spores found in *T. angulatus* specimens was consistent with that of *Myxidium* genus (Fig. 1).

3.1. *Myxidium imperatrizensis* n. sp. (Myxozoa: Bivalvulida)

3.1.1. Description of the taxon

Phylum Cnidaria Hatschek, 1888.

Class Myxosporaea Bütschli, 1881.

Order Bivalvulida Shulman, 1959.

Family Myxidiidae Thélohan, 1892.

Genus *Myxidium* Bütschli, 1882.

Species *Myxidium imperatrizensis* n. sp. (Fig. 1C).

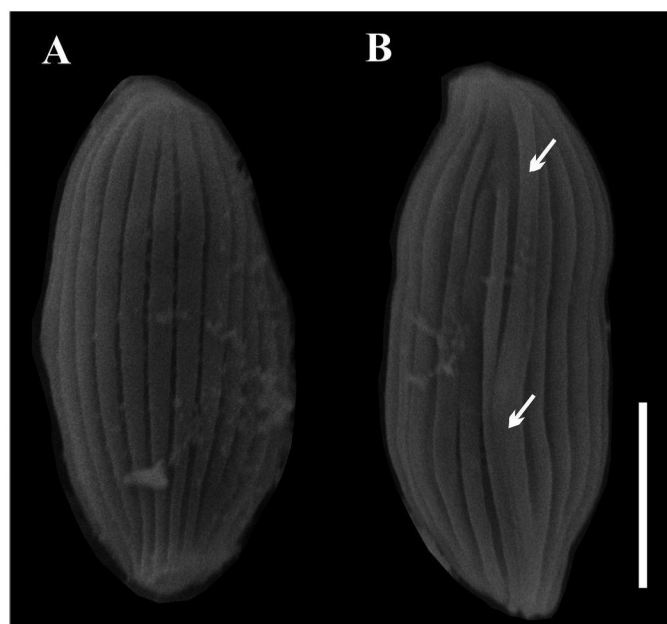


Fig. 2. Scanning Electron Micrography (SEM) of the spores of *Myxidium imperatrizensis* n. sp. (A – B). A – Valvular view; B– Sutural view; Arrows = Suture line. Scale bar = 10 μ m.

3.1.2. Taxonomic summary

Type host: *Triporthus angulatus* Spix & Agassiz, 1829 (Characiformes: Triportheidae).

Type locality: Tocantins River. Urban zone in the town of Imperatriz, in the eastern extreme of the Amazon biome, in Maranhão state, Brazil (5.5273°S, 47.4955°W).

Infection site: Spores dispersed within the gallbladder.

Prevalence: 8.6% of fish examined were infected.

Type material: Holotypes, in paraffin blocks, are deposited in the collections of the Carlos Azevedo Research Laboratory (LPCA/UFRA) in Belém, Brazil, and the Ecology and Limnology Laboratory (LEL/UEMA-SUL) in Imperatriz, Brazil. Slides stained with Hematoxylin-Eosin were deposited in the zoological collection of the National Institute of Amazonian Research (INPA) in Manaus, Brazil, under catalog number INPA 57. The partial sequence of the SSU rDNA gene of *Myxidium imperatrizensis* n. sp. was deposited in GenBank under accession number MN708206.

Spore morphology: *Myxidium imperatrizensis* spores n. sp. ($n = 30$) are elliptical, elongated perpendicular to the suture line, with an average length of $17.1 \pm 0.7 \mu$ m and width of $8.9 \pm 0.4 \mu$ m, with two slightly oval and symmetrical polar capsules, $6, 5 \pm 0.3 \mu$ m in length and $5.8 \pm 0.2 \mu$ m in width that open to opposite valves. The spiral spiral filament has seven to eight coils.

Etymology: the specific epithet is a reference to the city of Imperatriz in the state of Maranhão, Brazil, where specimens of *T. angulatus* infected with *Myxidium imperatrizensis* n. sp. were collected.

3.2. Morphological analysis

The features of the external topography of spore revealed by scanning electron microscopy, included crests and longitudinal grooves. While these features were more numerous in spores of *Myxidium*

Table 1
Morphological comparison of species of Myxidiidae parasites of fish in Brazil.

Taxon	Spore		polar capsules		Coils in polar filament	Host (order: family)
	length	width	length	width		
<i>Myxidium imperatrizensis</i> n. sp.	17.1 ± 0.7 (16.2–18.3)	8.9 ± 0.4 (8.1–9.5)	6.5 ± 0.3 (6–7)	5.8 ± 0.2 (5.5–6)	7–8	Characiformes: Triportheidae
<i>M. ceccarellii</i>	17.7 ± 0.5 (17.1–18.1)	10.4 ± 0.47 (9.8–11.3)	6.3 ± 0.5 (5.7–7.0)	6.4 ± 0.44 (5.7–6.9)	4–5	Characiformes: Anostomidae
<i>M. amazonense</i>	17.0 ± 0.9 (16.1–17.9)	3.7 ± 0.7 (3.0–4.4)	5.4 ± 0.5 (4.9–5.9)	3.4 ± 0.6 (2.8–4.0)	4–5	Siluriformes: Callichthyidae
<i>M. cholecysticum</i>	14.1 ± 1.1 (12.0–15.9)	7.8 ± 0.9 (6.4–9.6)	4.2 ± 0.9 (3.4–5.2)	3.6 ± 0.3 (3.0–4.1)	5–6	Characiformes: Characidae
<i>M. cruzi</i>	17–18	5–7	5–6	2–3	–	Characiformes: Triportheidae
<i>M. gurgeli</i>	14.6	8.5	3.4	3	–	Characiformes: Characidae
<i>M. striatum</i>	10–14	6–8	4	4	–	Perciformes: Sciaenidae
<i>M. volitans</i>	21.7 ± 0.3 (21.3–22.0)	5.6 ± 0.4 (5.2–5.9)	5.0 ± 0.4 (4.6–5.5)	2.3 ± 0.3 (2.0–2.5)	2–3	Scorpaeniformes: Dactylopteridae

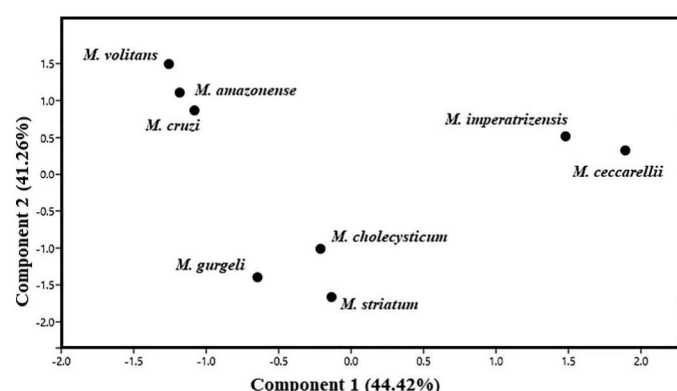


Fig. 3. Plot of the results of the Principal Components Analysis (based on the Hellinger distance) of the morphometric characteristics of the myxidiid parasites of Brazilian fish.

imperatrizensis n. sp. (Fig. 2) and the suture line is longitudinally sinuous in the spore body.

The comparison of the morphometry of the new species with other known myxidiids in Brazilian watersheds (Table 1), revealed a greater structural similarity with *Myxidium ceccarellii*, a parasite of characiform fish of the Anostomidae family that occur in the basin of the São Francisco River (Fig. 3). Based on the statistical tests of the broken-stick, only the first ordering axis showed a statistically significant relationship, with the observed clustering of the compared species being correlated mainly with the polar capsule morphometry.

3.3. Molecular and phylogenetic analyses

The amplification products of the SSU rDNA partial sequences had 1401 base pairs (bp) in *Myxidium imperatrizensis* n. sp. This species obtained the best BLAST score for the *M. ceccarellii* (KJ499821) sequence recovered from GenBank, with an identity of 82.3% and coverage of 97%.

The Bayesian phylogenetic inference was based on the two sequences obtained in the present study, combined with 40 closely related myxozoan taxa (BLAST identity ≥85%). These sequences included the two Brazilian *Myxidium* species available in GenBank, *M. amazonense*

(KT625442) and *M. ceccarellii* (KJ499821), which are the most morphometrically similar to the two new species described here. The basal clade of freshwater fish mixidid parasites included *Myxidium imperatrizensis* n. sp. and *M. ceccarellii*. (Fig. 4). *Myxidium amazonense* and the clade formed by other species of *Myxidium* and *Zschokkella* that infect freshwater hosts were more closely related to the clade composed by the species of *Zschokkella* that parasitize marine fish.

4. Discussion

Despite the considerable diversity of fish found in the tropical zone of South America [25,26], few data are available on the myxidiid parasites associated with these fish [2,4]. Although Myxidiidae is the second most diverse group of myxozoans in the world [1,10].

The present study describes one new species of myxozoan from the Myxidiidae family that parasitise the gallbladder of characiform fish collected on the middle Tocantins River, in the transition zone between the savanna environment of northeastern Brazil and the eastern end of the Brazilian Amazon Forest biome. This is only the second report of a species of *Myxidium* parasitising a triportheid fish in Brazilian continental waters [10], the first being the description of the occurrence of *M. cruzi* in the gallbladder of *Triportheus nematurus* in the Paraguay River basin [1,10]. The description of *Myxidium imperatrizensis* n. sp. it is the first report of the occurrence of parasites of this genus in fish of the species *Triportheus angulatus*, and it also represents an extension of the known geographical area of this genus of myxozoans to the Tocantins River basin.

The analysis of the morphological characteristics of the mixid species described here, together with the results of phylogenetic analyzes, supported the distinction of both *Myxidium imperatrizensis* n. sp. of the other mixidiids described. The increase in the number of genetic sequences, together with the use of spore morphological data and information about hosts, are necessary factors for a better understanding of the complex phylogenetic relationships between representatives of the myxidiidae family [5,6,27,28].

Declaration of Competing Interest

The authors declare that they have no conflicts of interest with regard to the publication of the present study.

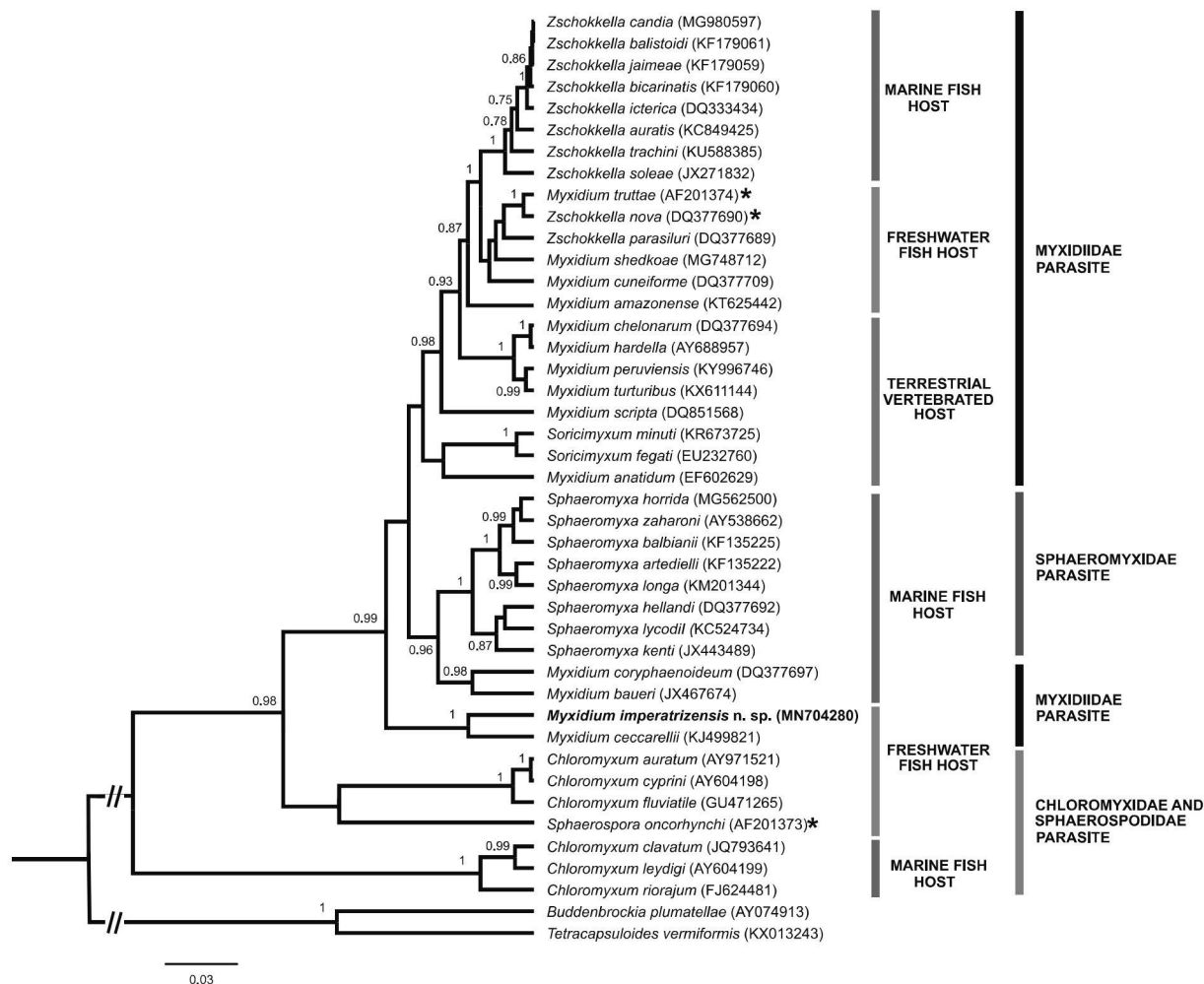


Fig. 4. Bayesian Inference analysis of the SSU rDNA gene of 41 myxozoans of the families Chloromyxidae, Myxidiidae, Sphaeromyxidae, and Sphaerosporidae. *Buddenbrockia plumatellae* and *Tetracapsuloides vermiformis* were included as the outgroup. The numbers indicate the posterior probability of the Bayesian analysis. The samples obtained in the present study are highlighted in bold type. Asterisk = parasite recorded in anadromous-catadromous host. The cut off in the tree represents a reduction of 20%.

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